

# WELCOME TO PROFESSIONAL CERTIFICATE PROGRAMME IN AGENTIC AI AND APPLICATIONS

**PROGRAMME ORIENTATION**  
**February 15, 2026**



# ORIENTATION - HOUSEKEEPING RULES

Use the Q&A box to drop a question to us.  
We'll stop regularly to answer these.

Drop a Quick Introduction with your Name and  
Experience

Join the session through your registered  
Canvas email\_id

Please keep yourself on mute.



# AGENDA

## 01. INTRODUCTION

- IITM Pravartak University Faculty
- Your Programme Leader
- Tech Certs Support team

## 02. PROGRAMME DETAILS

- Participant Profile
- Learning Journey
- Learning Platform

## 03. Q&A

- Next Steps



# INTRODUCTION



# INTRODUCTION: IITM FACULTY



**Prof. Madhusudhanan Baskaran**



- Principal AIML Consultant and Lead Faculty at IITM Pravartak
- Over 32 years across academia, industry, and government

## Expertise



- Designing AI agents that can plan, reason, and adapt autonomously
- Responsible and scalable AI systems leveraging LLMs, speech technologies, and document intelligence
- Leading initiatives in explainable AI and modular agent architectures
- Mentorship of deep-tech startups in India's AI transformation

## Publications



- The Scientific World Journal: Synchronous firefly algorithm for cluster head selection in WSN
- International Conference on Computer and Information Sciences: Improving security in Wireless Sensor Network using trust and metaheuristic algorithms
- Soft Computing (Vol. 24): Computerized grading of brain tumors supplemented by artificial intelligence

# INTRODUCTION: PROGRAMME LEADER



**Ajay K Singhal**

## **Experience and Background:**

- Senior Data & AI Engineering leader with 22+ years of experience, including strong recent focus on Generative AI and Agentic AI, delivering enterprise AI solutions across banking, retail, insurance, education, and engineering.

## **Core Skill sets:**

- Designing agentic applications using LangChain, LangGraph, and AutoGen with controlled tool invocation and validation.
- LLM integration, fine-tuning, RAG pipelines, vector databases, and multi-modal semantic search.
- Cloud-native AI engineering on AWS, Azure, and GCP using Docker, Kubernetes, and scalable orchestration.
- Secure AI architecture, benchmarking, inference optimization, and governance for production.

## **Major Projects :**

- Delivered LLM-powered agentic solutions integrated with enterprise data platforms, including benchmarking and response validation.
- Built secure GenAI workflows using Azure OpenAI and custom LLMs with controlled access boundaries.
- Implemented multi-modal semantic search using LLMs, ElasticSearch, and vector databases.



# TECH CERTS SUPPORT TEAM



**Asha**  
**[Associate – Academic Delivery]**



**Shiju**  
**[Tech Cert Support- Academic Delivery]**



**Hari**  
**[Escalation Manager – Academic Delivery]**

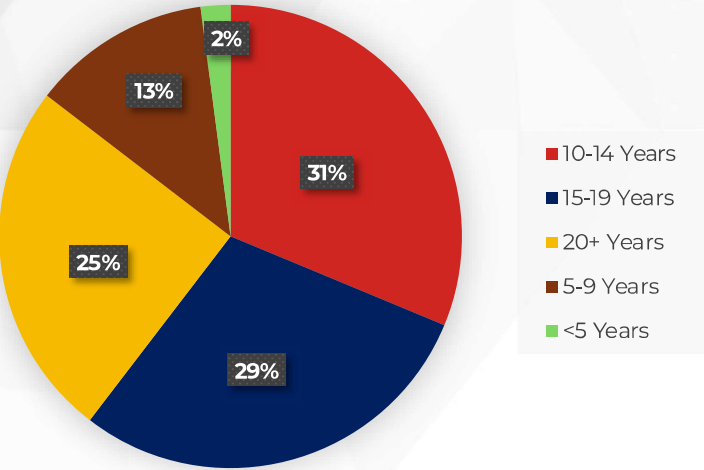
# PARTICIPANT DEMOGRAPHICS



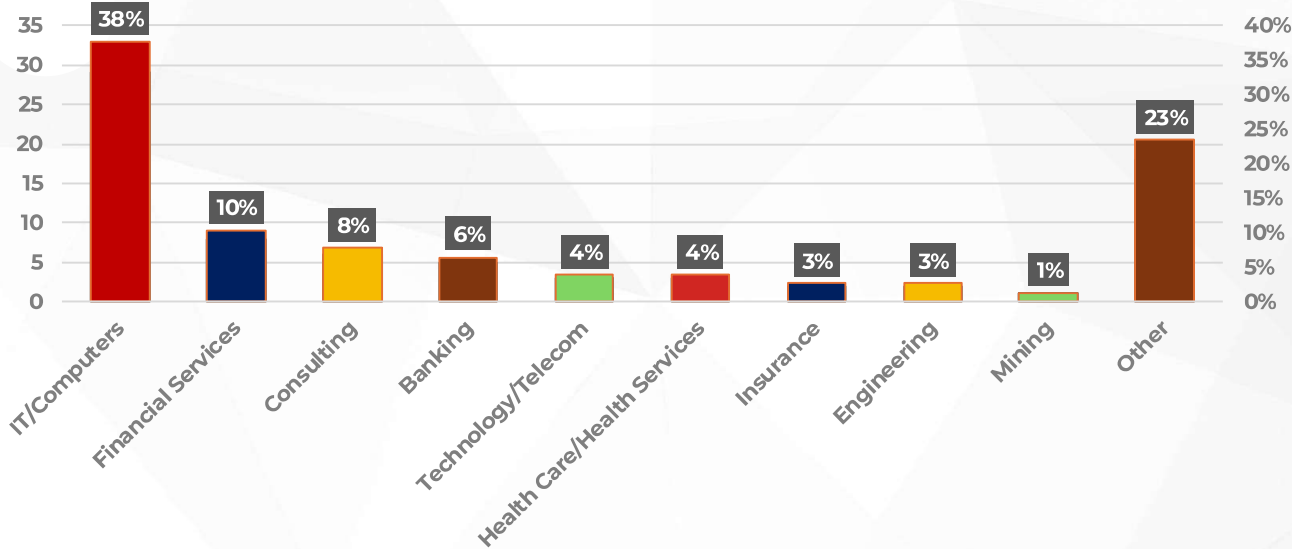


# PARTICIPANT DEMOGRAPHICS

Work Experience



Industry Domain



# PROGRAMME SUMMARY



# PROGRAMME SUMMARY

- **Programme Duration** : 5 months [Excluding Break weeks]
- **Eligibility Criteria** :
  - Minimum Graduate (10+2+3)
  - Diploma Holders with minimum 5 years of experience
  - Basic Math and Programming knowledge preferred
- **Learning Support** :
  - **Weekly Efforts** : 8 / 10 hours per week
  - **Pedagogy: Program is taught by Domain experts** through Weekly live sessions , 3 to 4 Faculty led Masterclasses(topics will be finalized by faculty & schedule will be shared before each masterclass).
  - **Campus Immersion** : 2 days optional campus immersion event at IIT Madras Research Park

# IITM FACULTY MASTERCLASS

- The program includes 3 to 4 Masterclasses conducted by IITM Pravartak faculty, focusing on emerging topics and practical applications in Agentic AI.
- The Masterclass topics will be finalized by the faculty, and the schedule will be shared in advance prior to each session.
- The Masterclass sessions will be conducted on weekdays, as weekends are completely dedicated to live sessions.

# PROGRAMME SUMMARY

- The programme includes initial dedicated weeks for learning and practicing programming to ensure you are ready for the technical modules that follow.
- If you have limited coding experience, these sessions will be critical to your success. If you are already comfortable with coding, they may feel introductory but will still provide valuable practice for the advanced modules ahead.
- Learners with less programming background may need to put in additional effort to keep pace.
- We encourage all learners to attend each of the programming live sessions to make sure they become well versed with coding which will help them complete later modules



# SAMPLE OF ASSIGNMENTS

The screenshot displays a file explorer on the left and a PDF viewer on the right. The file explorer shows a directory structure with folders 'voc' and 'data', and a 'work' folder containing two PDF files: 'Practice Assignment 1.2 Prompt Engineering and C...' and 'Solutions\_Assignment1.2\_Prompt\_or\_Refactor.pdf'. The PDF viewer shows the content of 'Solutions\_Assignment1.2\_Prompt\_or\_Refactor.pdf', which includes assignment prompts and a sample final solution in Python.

**Files**

- voc
- data
- work
  - Practice Assignment 1.2 Prompt Engineering and C...
  - Solutions\_Assignment1.2\_Prompt\_or\_Refactor.pdf

**Solutions\_Assignment1.2\_Prompt\_or\_Refactor.pdf**

Iteration 3 (testing + clarity):

*"Provide 3 asserts that validate the outputs on short inputs. Keep the implementation under 25 lines with clear comments."*

Sample final solution:

```
import re
from collections import Counter

def analyze_text(text, top_n=3):
    if not isinstance(text, str) or not text.strip():
        return {"word_count": 0, "char_count": 0, "top": []}
    cleaned = re.sub(r"[^A-Za-z0-9\s]", "", text).lower()
    tokens = [t for t in cleaned.split() if t]
    counts = Counter(tokens)
    return {
        "word_count": len(tokens),
        "char_count": len(text),
        "top": counts.most_common(top_n)
```

# SAMPLE OF ASSIGNMENTS

✓ Numpy Case Study Solution.ipynb ×

```
1 import numpy as np
2
3 # **Exercise 1: Creating and Manipulating NumPy Arrays**
4
5 # 1. Create a 1D NumPy array
6 arr = np.array([10, 20, 30, 40, 50])
7 print("1D Array:", arr)
8
9 # 2. Perform element-wise addition
10 arr_add = arr + 10
11 print("Array after adding 10:", arr_add)
12
13 # 3. Create a 2D NumPy array
14 arr_2d = np.array([[1, 2, 3], [4, 5, 6]])
15 print("2D Array:\n", arr_2d)
16
17 # 4. Indexing Exercise: Access and print the second row
18 second_row = arr_2d[1]
19 print("Second Row:", second_row)
```



[1]



Expand Output ▾

# PROGRAMME OUTCOMES

## With this programme, you will be able to:

- Build intelligent AI agents that can plan, reason, act, and adapt—just like a human teammate.
- Get hands-on with cutting-edge tools such as LangChain, CrewAI, and AutoGen to create real-world AI solutions across industries.
- Transform static AI into smart, goal-driven systems using GenAI, RAG, prompt engineering, and adaptive feedback loops.
- Deploy and optimize your own AI agents with memory, tools, and monitoring—ready for real users and real-world impact.
- Become a future-ready AI builder by solving practical problems through custom multi-agent workflows across business, tech, and beyond.

### Note :

**This is an application-oriented technical programme. Learners should expect coding, tool usage, workflow automation, and dedicate 8–10 hrs/week.**

**This programme not cover business theory, management frameworks, or strategy concepts**

**Some learners may need additional effort depending on their learning pace.**

# PROGRAMME JOURNEY



# YOUR LEARNING JOURNEY AT A GLANCE

## Your learning path

- **Live sessions:** Weekly live sessions by Programme Leader (**only live sessions and no recorded content**)
- **Foundation:** Python programming with practice sessions (ChatGPT support will also be used).
- **Core skills:** Designing agents, reasoning, memory, and multi-agent collaboration.
- **Advanced practice:** RAG systems, deployment, monitoring, and evaluation.
- **Integration:** Ethics, safety, governance, and the capstone project where you build and deploy your own agent.

## Projects & Assignments

- 20+ weekly assignments & mini-projects
- All build toward final **capstone project** (design & deploy a functional AI agent)

# PROGRAMME JOURNEY

| Module No | Duration (weeks) | Module Title                          | Learning Format                                            |
|-----------|------------------|---------------------------------------|------------------------------------------------------------|
| 1         | 1                | Getting Started with Python & ChatGPT | Live Session with Programme Leader + Self check activities |
| 2         | 1                | Data Types, Variables & Control Flow  |                                                            |
| 3         | 1                | Functions, & Working with Libraries   |                                                            |
| 4         | 1                | Fundamentals of AI & ML               |                                                            |
| 5         | 1                | Large Language Models (LLMs)          |                                                            |
| 6         | 1                | Embedding Models & Vector Basics      |                                                            |
| 7         | 1                | Agentic Tools in Python               |                                                            |
| 8         | 1                | Introduction to Agentic AI            |                                                            |



# PROGRAMME JOURNEY

| Module No                                          | Duration (weeks) | Module Title                                  | Learning Format                                                  |
|----------------------------------------------------|------------------|-----------------------------------------------|------------------------------------------------------------------|
| IBM : Generative AI for Business and Professionals |                  |                                               | Live Session with<br>Programme Leader + Self<br>check activities |
| 9                                                  | 1                | Programming & Frameworks for Agentic Systems  |                                                                  |
| 10                                                 | 1                | Agent Architectures & Collaboration           |                                                                  |
| 11                                                 | 1                | Decision-Making & Planning in Agents          |                                                                  |
| 12                                                 | 1                | Memory & Knowledge Retrieval in Agents        |                                                                  |
| 13                                                 | 1                | Prompt Engineering & Adaptive Instructions    |                                                                  |
| 14                                                 | 1                | Learning & Adaptation in Agents               |                                                                  |
| 15                                                 | 1                | Advanced Retrieval-Augmented Generation (RAG) |                                                                  |

# PROGRAMME JOURNEY

| Module No                                       | Duration (weeks) | Module Title                              | Learning Format                                                  |
|-------------------------------------------------|------------------|-------------------------------------------|------------------------------------------------------------------|
| IBM : Building AI Agents with RAG and LangChain |                  |                                           | Live Session with<br>Programme Leader + Self<br>check activities |
| 16                                              | 1                | Deploying & Monitoring Agentic Systems    |                                                                  |
| 17                                              | 1                | Agent Evaluation & Debugging              |                                                                  |
| 18                                              | 1                | Ethics, Safety & Governance in Agentic AI |                                                                  |
| IBM: Responsible and Ethical Generative AI      |                  |                                           |                                                                  |
| 19                                              | 1                | Real-World Applications & Case Studies    |                                                                  |
| 20                                              | 1                | Low-Code Tools Deep Dive                  |                                                                  |
| 21                                              | 1                | Capstone Project – Build Your Own Agent   |                                                                  |

# TOOLS USED

| Tools                           | Mapping with Modules     |
|---------------------------------|--------------------------|
| Flowise (demo)                  | 8, 9, 13, 15, 19, 20, 21 |
| Autogen Studio (demo)           | 8, 20                    |
| Google Teachable Machine (demo) | 4                        |
| Scikit-learn (visual)           | 4                        |
| TensorFlow Playground           | 4                        |
| LangFlow                        | 9, 20                    |
| OpenAgents (as needed)          | 9 (optional), 19, 20     |
| HuggingFace Spaces              | 5, 16, 19                |
| LangChain (prompt templates)    | 5, 13                    |
| Tokenizer Visualizers           | 5                        |
| CrewAI                          | 9, 10, 11, 19, 21        |
| Autogen                         | 9, 10, 11                |
| LangChain Agents                | 10, 11                   |
| AgentVerse (optional)           | 10                       |
| LangChain Agents (PlannerAgent) | 11                       |
| Autogen Planning Loops          | 11                       |
| ChromaDB                        | 6, 12, 15, 21            |
| Pinecone                        | 12, 15, 21               |
| LangChain Memory                | 12                       |
| FAISS                           | 6, 12, 15, 21            |
| OpenAI Embeddings               | 6, 12, 21                |
| OpenAI Playground               | 5, 13                    |
| LangChain PromptTemplates       | 13                       |

| Tools                                | Mapping with Modules                           |
|--------------------------------------|------------------------------------------------|
| PromptLayer (optional)               | 13, 17                                         |
| OpenAI API (with temperature tuning) | 14, 21                                         |
| LangChain + feedback loops           | 14                                             |
| RL demo environments (e.g., CleanRL) | 14                                             |
| simple Gridworld visual              | 14                                             |
| LangChain RAG Chain                  | 15                                             |
| ChromaDB / Pinecone                  | 15                                             |
| Hugging Face Transformers            | 6, 15, 21                                      |
| Streamlit                            | 16, 19, 21                                     |
| LangChain + FastAPI                  | 16                                             |
| Render / Replit                      | 16                                             |
| LangSmith                            | 16, 17                                         |
| PromptLayer                          | 13, 17                                         |
| Autogen logs                         | 17                                             |
| LangChain Tracing                    | 17                                             |
| Gradio logs                          | 17                                             |
| OpenAgents                           | 19, 20                                         |
| LangChain Hub                        | 19, 20                                         |
| CrewAI templates                     | 19                                             |
| Streamlit demos                      | 19                                             |
| LangChain                            | 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17, 21 |
| OpenAI                               | 7, 21                                          |
| Hugging Face                         | 5, 6, 15, 16, 21                               |

- Given the dynamic nature of this program, the tools may be subject to change or alteration to best suit the outcomes of each module.
- Tools will be provided via virtual labs for learning, as per the curriculum. Access will be given when the respective modules are taught.

# A TYPICAL WEEK



## ACADEMIC MODULE RELEASE

- Self-check activities
- 1 practice quiz and 1 discussion prompt each week
- Required assignments or Mini projects
- Slide/transcript downloads



## LIVE SESSIONS

- Program Leader (Domain expert) conduct interactive live sessions along with hands-on (**up to 3 hours**)
- Live session would be on **Saturday & Sunday**
- 3-4 Masterclasses with IITM Faculty

# CODE OF CONDUCT

## PROGRAMME AGREEMENT

- Respect the privacy of other participants
- Respect the diversity of opinions and cultures
- Use a single account on the platform and provide correct information about yourself
- Do not reproduce content from your course or from other participants unless expressly allowed under the copyright terms by the programme administrator
- Do not submit the work of others as your own or engage in any activity that will improve or harm others' points
- If you believe that someone has violated this agreement, please contact [techcertsupport@emeritus.org](mailto:techcertsupport@emeritus.org).

## SUBMISSION

- Assignments submitted on time will be reviewed within two week of submission due date.
- If an assignment is not submitted by the due date, there would be no opportunity to submit it on a later date.
- There would be course end assignment at the end of the course which would be proctored and time bound.
- The assignments of the course can be in the form of Quizzes, Participation , Individual Assignment.

## ACADEMIC INTEGRITY

- Participants are individually responsible for work they submit. They must not submit the work of others as their own and do not engage in any activity that will improve or harm others' scores.
- Plagiarism is a form of academic dishonesty.
- Usage AI tools are beneficial but should supplement, not replace, critical thinking.
- Plagiarism detected by AI, including ChatGPT, results in marked-down assignments.
- Proper citations should be given for any reference used for assignments

# COMPLETION REQUIREMENTS

- **Achieve a minimum average score of 70% across all mandatory graded assignments.**
- **Successfully complete the capstone project.**
- **Maintain at least 50% attendance in live sessions** to be eligible for programme completion.





# FINAL CERTIFICATE CRITERIA

To receive your Digital Certificate of Completion, you must:

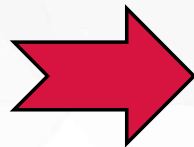
**Maintain 50% Attendance  
(Live Session + Faculty Led  
Masterclasses)**



**Complete all Graded  
assignments and achieve a  
minimum overall score of  
70%**



**Successfully complete 1  
Capstone Project**



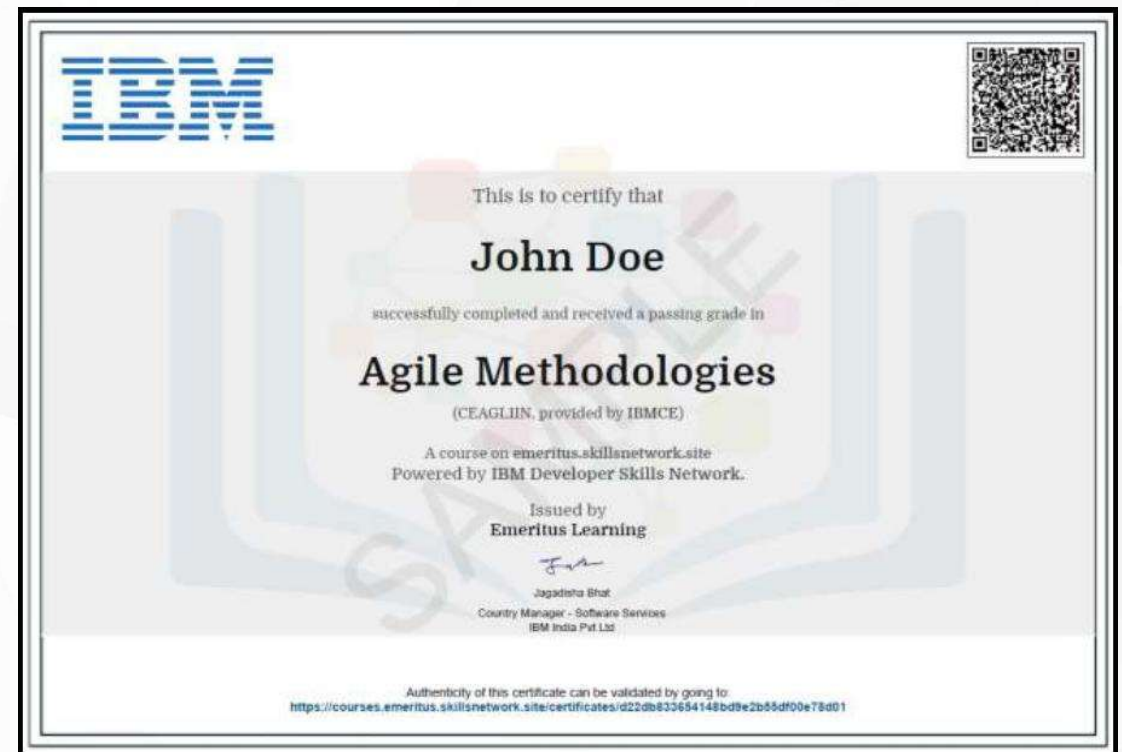
# IBM COURSES

We are pleased to announce that three IBM courses will be introduced. Platform access will be provided during the designated week, and participants will be required to complete the assigned tasks in order to unlock the IBM certificate.

Generative AI  
for Business  
and  
Professionals

Building AI  
Agents with  
RAG and  
LangChain

Responsible  
and Ethical  
Generative AI



# CAREER SERVICES BENEFITS

You'll get the access to the **Career prep videos after completion**, and **IIMjobs Pro support** will remain available for **six months** post-program.

## Career Services (CS Modules)

### Recorded videos on:

Resume and Cover Letter

These recorded videos will cover the basic elements of building the correct resume and cover letter to attract the right opportunities.

### LinkedIn Profile and Navigating the Job Search

These recorded videos will help you understand why LinkedIn is an essential personal branding tool and how to utilize it for professional networking, job search and more.

### Interview Preparation

These recorded videos will impart an overview of the branding concept which will help students communicate their value and unique assets to others during the interview process

# CAREER SERVICES BENEFITS

## Career Services (IIMJobs Pro)

### 6 months IIMJobs pro subscription

- **Priority Application Access** – Get your profile highlighted to recruiters and jump the queue.
- **Exclusive Jobs** – Unlock premium job postings not visible to free users.
- **Resume Visibility Boost** – Your profile ranks higher in recruiter searches.
- **Job Recommendation Insights** – Get personalized job suggestions based on your profile and goals.
- **Early Access to Roles** – Apply before positions go live to general users.

# CANVAS WALKTHROUGH



# CANVAS WALKTHROUGH



My Programs

My Profile

Calendar

Notification Settings

Inbox

Settings

Policies and Terms

My Programs

Active Programs

Completed Programs

IIT MADRAS

IITM Pravartak - Professional Certificate Programme in Agentic AI and Applications - February 2026 [Parent Instance]

Starting Soon

Starts on 02/11/2026

Go to Program

IITM Pravartak - Professional Certificate Programme in Agentic AI and Applications - February 2026

In Progress

Go to Program

IITM

PRAVARTAK

CATALYSING INNOVATION

A Technology Innovation Hub of IIT Madras

# CANVAS WALKTHROUGH

How to update your time under your Canvas

**EMERITUS**

- My Programs
- My Profile
- Calendar
- Notification Settings
- Inbox
- Settings**
- Policies and Terms

### Test Student's Settings

Full Name: \* This name will be used for grading.

Display Name: People will see this name in discussions, messages and comments.

Sortable Name: This name appears in sorted lists.

Pronouns: None This pronoun will appear after your name when enabled

Language: System Default (English (United States))

Time Zone: Mumbai

Ways to Contact

Email Addresses

+ Email Address

Other Contacts Type

+ Contact Method

Edit Settings

Download Submissions

Time Zone: Mumbai (+05:30)

☐ I want to receive information, news and tips from instructors

Cancel Update Settings



# CANVAS WALKTHROUGH

  
IITM PRAVARTAK - PROFESSIONAL  
CERTIFICATE PROGRAMME IN...

- Home
- Modules
- People
- Vocareum
- Grades
- Live Sessions
- Discussions
- Announcements 0
- Support

## Recent Announcements



Programme Leader Introduction, Course Outline and Certificate Criteria

Dear Participant,Welcome to the Professional Certificate Programme in Agentic AI and Applications.We are delighted to introduce Arul...

Posted on:

Feb 3, 2026, 7:39 AM

## IITM Pravartak - Professional Certificate Programme in Agentic AI and Applications - February 2026

Start Here

Programme Outline

Completion Requirements



# CANVAS WALKTHROUGH

The screenshot shows the Canvas EMERITUS user interface. On the left is a sidebar menu with options: My Programs, My Profile, Calendar (highlighted), Notification Settings, Inbox, Settings, and Policies and Terms. The main area displays a calendar for February 2026. At the top of the calendar, there are navigation buttons: 'Today', left and right arrows, the month 'February 2026', and tabs for 'Week', 'Month' (selected), and 'Agenda'. The calendar grid shows days of the week (SUN to SAT) and dates. An event titled '1:30p Programme ...' is scheduled for Saturday, February 14th. Another event titled '1:30p Live Session...' is scheduled for Saturday, February 21st. The date February 11th is highlighted in grey.

## Sync to your personal calendar!

- Select Calendar from the Canvas menu to see in your programme calendar listing all events and due dates. You can sync this with your personal calendar:

- [How to subscribe to Calendar feed using personal Outlook calendar](#)
- [How to subscribe to Calendar feed using Google calendar](#)

# ATTENDANCE GUIDELINES

**Use Registered Credentials:** Participants must join sessions using their registered Name and Email ID exactly as provided during registration.

**Attendance Verification:** Attendance is valid only when session details match registration records to ensure genuine participation.

**No Exceptions Allowed:** Queries about attendance won't be accepted if guidelines are not followed, ensuring fairness and transparency.

**Double-Check Details:** Learners should verify their login details (Registered Name and Email ID) before joining to avoid attendance issues and disruptions.



# NEXT STEPS



# NEXT STEPS

## ORIENTATION MODULE

Complete all the materials in the orientation module.

## Support CHANNEL

[techcertsupport@emeritus.org](mailto:techcertsupport@emeritus.org)

## CALENDAR

Mark your calendars for next week's live session  
February 21, 2026 10:00 AM IST

## REFERENCES

Introduce on "Meet your Fellow Learners"

# SUPPORT CHANNELS



**Support ID:** [techcertsupport@emeritus.org](mailto:techcertsupport@emeritus.org)

Please allow us 24-48 hours to respond.

# LIVE SESSION EXPERIENCE SURVEY

**Before we proceed to Q&A  
Take 2 minutes to share your  
feedback with Us!**





# Q&A



**THANK YOU!!!**